PHASE 1 REPORT: DATA COLLECTION AND PREPARATION

Ezgi Dönmez · DS588

1. INTRODUCTION

In Phase 1, my goal was simple to state but took up most of my time: get clean, reproducible daily prices for Turkish funds and US-listed ETFs, and remove only the instruments that cannot be used, each for a documented reason. I did not hand-pick any assets. The eligible universe has 4,908 assets, all saved as documented CSV files for Phase 2.

I chose a study window of 1 April 2026 to 1 October 2026 (both dates included). Six months is not too long, but it is long enough to compute weekly returns, volatility and correlations, and it kept the manual Turkish downloads manageable. I plan to use weekly rolling steps in the validation phase because of this short window.

2. DATA SOURCES

US ETFs. I wrote a small script, `abd_etf_download.py`, that first reads the official Nasdaq Trader symbol lists (`nasdaqlisted.txt` and `otherlisted.txt`) and keeps only rows marked as ETF and not test issues (Nasdaq Trader, n.d.). It then downloads daily prices from Yahoo Finance through `yfinance` (Aroussi, n.d.; Yahoo Finance, n.d.) in batches of 20 tickers with a 3-second pause, so it does not hit rate limits. If the run is interrupted, it saves what it already has and continues from the missing tickers next time. Every row stores the date, ticker, fund name, currency (USD), source and a `retrieved_at_utc` timestamp, which acts as my frozen snapshot. The result was **5,746 ETFs and 676,639 rows**.

Turkish funds (TEFAS). I first tried to pull TEFAS data through its API, but this did not work well. Repeated requests started to time out, and each request only allowed about one month of data, so collecting six months for every fund was not practical. Instead, I downloaded the data manually from the TEFAS website (TEFAS, n.d.): for each month from April to the start of October I selected all funds and exported the full data, once for exchange-traded funds (*borsa*) and once for securities investment funds (*menkul*). This gave me 12 CSV files (6 months x 2 fund types), stored under `Data/Raw/tr`. The September files run until 1 October, so the whole window is covered.

I also downloaded the pension funds (*emeklilik yatırım fonları*, BES) from TEFAS at first, but later deleted them: these funds can only be held inside a private pension contract and cannot be bought or sold freely like normal investment funds, so they are outside the scope of this project.

Exchange rate. Since the portfolio mixes TRY and USD assets, the notebook downloads USD/TRY (`USDTRY=X`) from Yahoo Finance once and saves it. Over the window it moved from 44.45 to 49.03.

Source	Instruments	How I got it	Price I use	Currency
TEFAS, *borsa* exports	BYF (exchange-traded funds)	Manual monthly export	Borsa İstanbul bulletin price	TRY
TEFAS, *menkul* exports	YAT (mutual funds)	Manual monthly export	Unit price (NAV)	TRY

Source	Instruments	How I got it	Price I use	Currency
Yahoo Finance	US-listed ETFs	abd_etf_download.py	Adjusted close	USD
Yahoo Finance	USD/TRY rate	Notebook, saved once	Daily close	TRY per USD

Keeping BYFs and mutual funds apart was important, because they trade differently. A BYF trades on Borsa İstanbul like a stock, so its bulletin price is the price you can actually trade at. A mutual fund is bought through TEFAS at its unit price, which is set after you send the order (forward pricing).

3. DATA CLEANING AND QUALITY CHECKS

All cleaning is done in one notebook, `phase1_preparation.ipynb`, and every rule is a named setting at the top so it is easy to change and rerun.

Reading the TEFAS files. The exports start with a few info lines before the table, so the reader searches for the `Fon Kodu,` header line. TEFAS uses a decimal comma, which I convert to a dot. Dates are parsed strictly as `DD.MM.YYYY`; an earlier version of my code used a mixed format, which silently swapped day and month for some dates, so I fixed this on purpose. In total the 12 files gave 255,138 rows and 2,091 funds. I also checked that each monthly file has a similar number of funds (about 30 BYFs and 2,000 mutual funds per month), so none of the exports was missing funds.

Removing bad rows. There were no duplicate (fund, date) rows. I removed 982 rows with a missing or zero price (41 funds affected), leaving 254,156 clean TR rows. The US file had no duplicates and no invalid prices.

Adjusted prices without double-counting. For US ETFs I use `adj_close`, which already includes dividends and splits, so its daily change is the total return. Because of this I didn't add the `dividends` column again. I keep `close` and `dividends` only for documentation. TEFAS unit prices already reflect the fund's income, so no extra adjustment was needed there.

Trading calendars. The two markets have different holidays. Turkey had 125 trading days (missing days such as 23 April, 1 May, 19 May and the 27 to 29 May holiday), and the US had 127 (for example 25 May, 19 June, 3 July and 7 September). Only 120 days are common. I decided not to fill these gaps, because filling a holiday with the previous price would create fake 0% returns and distort volatility and correlation.

Suspicious jumps. I flagged any TR fund moving more than 25% in one day and any US ETF moving more than 40%, since such moves usually mean a data problem (a price reset or split) rather than a real market move. This flagged 63 TR funds and 137 US ETFs.

4. ELIGIBILITY CONTROL

After cleaning, I applied a short list of eligibility filters. Each filter removes assets that cannot be used, for one written reason: access rules, the project's own rules, missing or broken data, or no real trading. Nothing is removed for being small, unpopular or performing badly. Out of 2,085 Turkish funds, **707 are eligible**, and out of 5,746 US ETFs, **4,201 are eligible**.

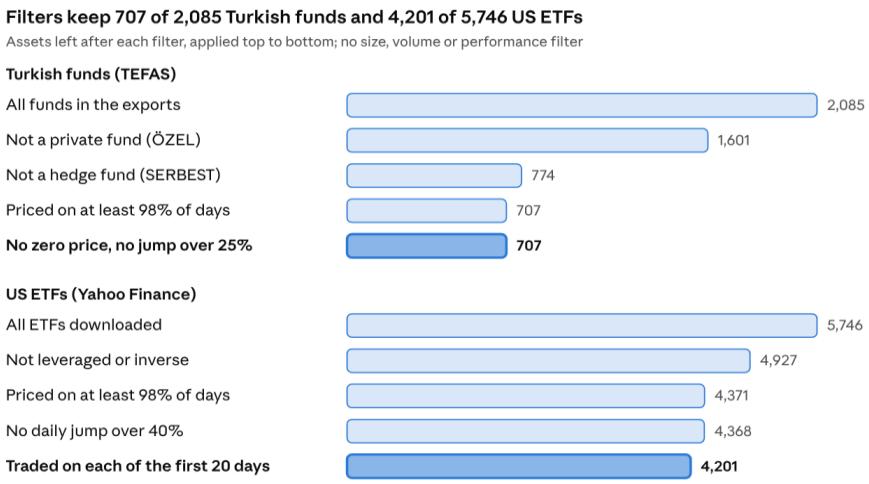


Figure 1. Eligibility funnel: number of Turkish funds and US ETFs remaining after each filter (1 April to 1 October 2026).

For Turkey, the biggest cut comes from private funds (*ÖZEL FON*) and hedge funds (*SERBEST FON*). These are only open to specific or qualified investors (SPK, 2013), so a normal investor cannot buy them, and removing them takes the list from 2,085 to 774 funds. Next, I drop funds that do not have a price on at least 98% of trading days (for example, a fund launched during the window or missing from one of the exports), because they cannot be simulated over the whole window. This leaves 707 funds. The checks for zero prices and for one-day jumps above 25% removed no further funds. I didn't filter by fund size: a small fund is still investable, so size is kept only as a description.

For the US, I first remove 819 leveraged and inverse ETFs. The project allows no short selling or borrowing: a 2x or 3x fund is the same as buying the index with borrowed money, and an inverse ("Short", "Bear") fund is the same as shorting it. The Nasdaq lists have no leverage flag, so they are detected from the fund name. Actively managed long/short equity funds stay in, because their aim is not a multiple or the opposite of an index. Next, 556 ETFs fail the 98% coverage rule (554 of them were launched during the window) and 3 have a one-day jump above 40%. Finally, 167 ETFs did not trade (zero volume) on at least one of the first 20 trading days. For those, the closing price is not a price anyone actually traded at, so it cannot be used for a realistic simulation.

Each step and its reason is saved in `data_quality_report.csv`, and every excluded asset carries the name of the step that removed it in `instruments.csv`.

The universe is every asset that passes the filters above: 707 Turkish funds plus 4,201 US ETFs, 4,908 assets in total. Deciding which of them to hold is the job of the Phase 2 algorithm.

5. OUTPUTS, TRADING ASSUMPTIONS AND LIMITATIONS

The notebook writes everything to `Data/Processed/`: `prices_long.csv` (one row per date and eligible asset), `instruments.csv` (every asset with its metadata, quality flags, eligibility and the reason it was excluded), `universe.csv`, `universe_prices_wide.csv` (4,908 assets x 132 dates, gaps not filled), `fx_usdtry.csv` and `data_quality_report.csv`. At the end, nine validation checks run automatically (no duplicates, positive prices, dates inside the window, both markets present, FX available, and so on), and all of them passed. Every column of every file is defined in my data dictionary (DATA_DICTIONARY.md), and the README explains how to reproduce the raw data and the outputs.

The project also asks not to assume that every fund can be traded instantly at the observed price, so I recorded the following assumptions for the simulation:

Type	Where it trades	Price	Settlement (assumed)
US ETF	NYSE / Nasdaq, intraday	Adjusted close	T+1
BYF	Borsa İstanbul, intraday	Bulletin closing price	T+2
TEFAS mutual fund	TEFAS order, forward pricing	Unit price	Varies by fund (often T+0 to T+3)

These are working assumptions. The exact settlement days of each fund are listed on its TEFAS and KAP pages (KAP, n.d.) and will be applied in the simulation.

Limitations

- Short history. Six months of data is enough for weekly signals, but not for signals that need a long look-back.
- Survivorship bias. The data only contains funds that still exist today. Funds that closed before October 2026 are missing, which can make past results look slightly better than they really were.
- Labels from fund names. Turkish fund categories and US leveraged or inverse ETFs are identified from fund names, because there is no official flag for them. I checked the matched names against the full lists, but a few funds may still be labelled wrongly.
- Delayed prices for Turkish funds that hold foreign stocks. These funds set today's price using yesterday's US closing prices, so their daily returns follow the US market with a one-day delay (Chalmers et al., 2001). In my data, AFT and YAY have a correlation of 0.52 and 0.59 with the previous day's QQQ return, but only about 0.05 with the same day's. Weekly returns largely remove this problem.

Next step (Phase 2)

I will start from universe_prices_wide.csv. Because the two markets have different holidays, I will first turn the daily prices into weekly prices (the last available price of each week), so both markets line up on the same dates. Then I will build the price-only algorithm that decides which eligible assets to hold and with what weights.

Use of generative AI

I used Claude (Anthropic, 2026) as a support tool for drafting and reviewing parts of the code and text. The data collection, the design decisions and the final content are my own: I directed the work, corrected the outputs where needed, ran all the code myself and checked every result against the notebook outputs.

References

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